
Airline Operations Data Warehouse - Design, ETL & Analytics

Airline Dataset Operations Analysis

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1. Introduction

1.1 Background

The global aviation industry serves as a critical artery of the world economy, connecting passengers, freight, and commerce across continents and time zones. According to IATA's 2025 analysis, worldwide passenger demand - measured by Revenue Passenger Kilometres (RPK) - grew by 5.3% compared to the previous year, slightly outpacing capacity growth of 5.2% as measured by Available Seat Kilometres (ASK). This imbalance produced an unprecedented Passenger Load Factor (PLF) of 83.6%, the highest annual utilisation rate ever recorded [1].

Within this context, international routes expanded at a robust 7.1% year-on-year, whilst domestic markets grew at a more modest 2.4%. Asia Pacific carriers achieved the second-highest growth rate globally at 7.8%. The Australian domestic market saw passenger traffic grow by approximately 2.9%, while capacity expansion drove a 2.8 percentage-point decline in utilisation, bringing the load factor to 82.2% [1].

Despite strong global performance, flight disruptions - including delays and cancellations - continue to impose high operational and reputational costs on airlines and airports. Understanding the patterns behind these disruptions is therefore of pressing interest to governments, airlines, and passengers alike.

1.2 Purpose of the Data Warehouse

This project develops a data warehouse for airline operations analysis, integrating passenger demographics, flight performance outcomes, geographic location, and time-based data. The warehouse is designed to support:

- Identification of patterns in flight delays and cancellations across time and geography
- Multi-dimensional analysis of flight performance at varying levels of granularity
- Data-driven insights to assist governments and airline operators in improving service delivery
- Evidence-based recommendations for civil aviation development policy

1.3 Dataset Overview

The primary dataset is the Airline Dataset (Updated v2) sourced from Kaggle [2], comprising records of individual passenger flight events. It includes passenger demographics (age, gender, nationality), operational details (airport, country, continent, departure date, flight status), and pilot information. Supplementary reference data from OurAirports include datasets on airports, countries, regions, and navigational aids [3].

One data quality issue was identified: the Arrival Airport column contains the IATA code of the departure airport rather than a genuine destination field — an ETL transformation error in the source pipeline. This is addressed in the pre-processing stage (Section 5).

2. Dimensional Modelling (Kimball's Four Steps)

The data warehouse was designed following Kimball's four-step dimensional modelling methodology, which provides a structured approach to transforming operational data into an analytical structure suitable for decision support [4].

2.1 Step 1 - Business Process

The business process modelled is flight operations and passenger movement. Each record in the source dataset represents a discrete flight event involving an individual passenger at a specific airport on a given date. This captures the operational lifecycle of passenger-facing flight activities, encompassing departures, scheduling outcomes (on-time, delayed, or cancelled), and the demographic characteristics of the travelling population.

2.2 Step 2 - Grain

The grain is defined as one record per passenger per flight event. This is the finest level of detail available in the dataset and ensures maximum analytical flexibility. Choosing a fine-grained approach allows the warehouse to answer both highly specific queries (e.g., the cancellation rate for adult passengers departing from European airports on weekends) and broad aggregate queries (e.g., overall monthly flight volumes by continent).

Selecting too coarse a grain - for instance, aggregating at the airport or monthly level — would have pre-empted many analyses of interest and limited the usefulness of the warehouse for passenger-level demographic queries [4].

2.3 Step 3 - Dimensions

Four dimensions were identified to support the range of business queries described in Section 3. Each dimension captures a meaningful analytical axis and includes a concept hierarchy enabling roll-up and drill-down operations [4].

Dimension	Table Name	Concept Hierarchy
Date	dim_date	Day → Month → Quarter → Year (+ Weekend Flag)
Location	dim_location	Airport → Country → Continent
Passenger	dim_passenger	Nationality → Age Group / Gender
Flight	dim_flight	Flight Status (On Time / Delayed / Cancelled)

Table 1 — Dimension summary with concept hierarchies

The Date dimension includes a Weekend Flag as a derived attribute, enabling analysis of whether weekend travel correlates with different flight outcomes. The Passenger dimension incorporates both raw Age and a derived Age_Group attribute (Young: 0-18, Adult: 18-59, Senior: 60+), providing flexibility for demographic roll-up queries. The Location dimension reflects the geographic hierarchy of OurAirports reference data, enabling aggregation from airport-level to continental-level [3].

2.4 Step 4 - Measures (Fact Table)

The fact table, fact_flight, stores one row per passenger flight event. The measures and foreign keys are as follows:

Column	Type	Description
flight_count	Measure	Always 1; used for COUNT aggregations
delay_flag	Measure	1 if Flight Status = Delayed, else 0
cancel_flag	Measure	1 if Flight Status = Cancelled, else 0
date_id (FK)	Dimension Ref	References dim_date
location_id (FK)	Dimension Ref	References dim_location
passenger_key (FK)	Dimension Ref	References dim_passenger
flight_id (FK)	Dimension Ref	References dim_flight

Table 2 — Fact table: fact_flight

3. StarNet Diagram and Query Footprints (Q1–Q7)

The StarNet diagram provides a visual representation of the multi-dimensional analysis space, illustrating the dimensions and their associated concept hierarchies that the data warehouse can navigate [5]. The four axes of the StarNet are:

- LOCATION: Airport → Country → Continent → All
- DATE: Day → Month → Quarter → Year → All (Weekend Flag at the Day level)
- PASSENGER: Nationality → Gender / Age Group → All
- FLIGHT: Status → All

Each business query is represented as a footprint on the StarNet — a polygon whose vertices indicate the specific level of each dimension at which the query operates. Seven queries were designed to demonstrate the analytical capabilities of the warehouse, covering country demand, temporal trends, weekend patterns, airport performance, passenger demographics, nationality analysis, and cancellation rates.

Q1 -Country-level Passenger Demand

Business question: Which countries generate the highest volume of flight events across all years and passenger types?

This query aggregates flight_count across all time periods and all passengers, drilling into Location at the Country level. It enables governments and airline planners to identify which national markets drive the most traffic, informing route planning and airport investment decisions.

StarNet footprint: Location anchors at Country; Date, Passenger, and Flight are rolled up to All.

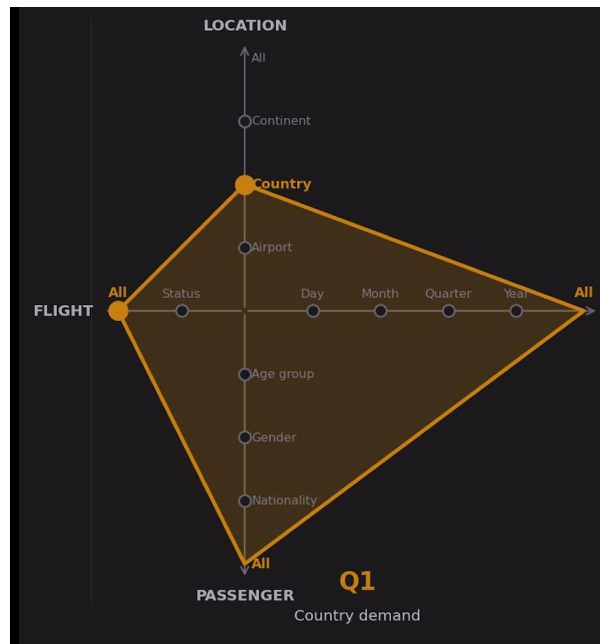


Figure 1 — Q1 StarNet Footprint: Country Demand

Q2 — Monthly Flight Volume Trends

Business question: How does the total number of flight events vary month by month across all locations and passenger types?

Drilling into the Date dimension at Month level whilst rolling Location and Passenger to All, this query reveals seasonal patterns in flight volume. Identifying peak travel months is essential for capacity planning, staffing, and airline scheduling optimisation.

StarNet footprint: Date anchors at Month; Location, Passenger, and Flight are at All.

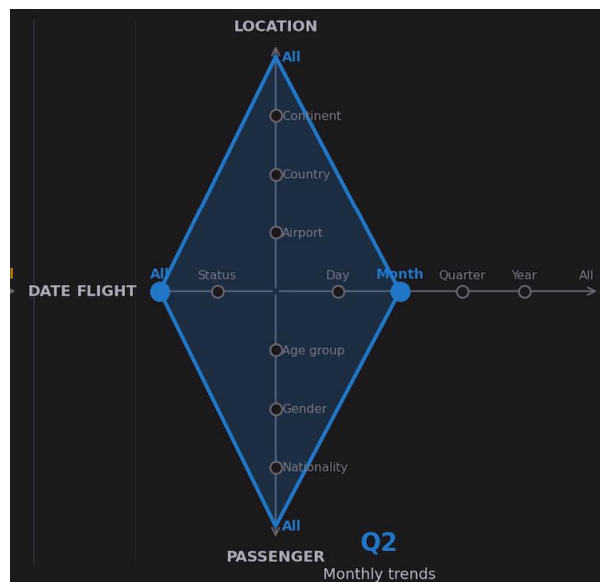


Figure 2 — Q2 StarNet Footprint: Monthly Trends

Q3 — Weekend vs Weekday Flight Patterns

Business question: Is there a measurable difference in flight volumes or outcomes between weekday and weekend departures?

This query uses the derived Weekend Flag at the Day level of the Date dimension, rolling Location and Passenger to All. It captures the operational contrast between leisure-dominated weekend travel and business-dominated weekday travel, with implications for pricing strategy and resource allocation.

StarNet footprint: Date anchors at Day (Weekend Flag); Location and Passenger are at All; Flight at All.

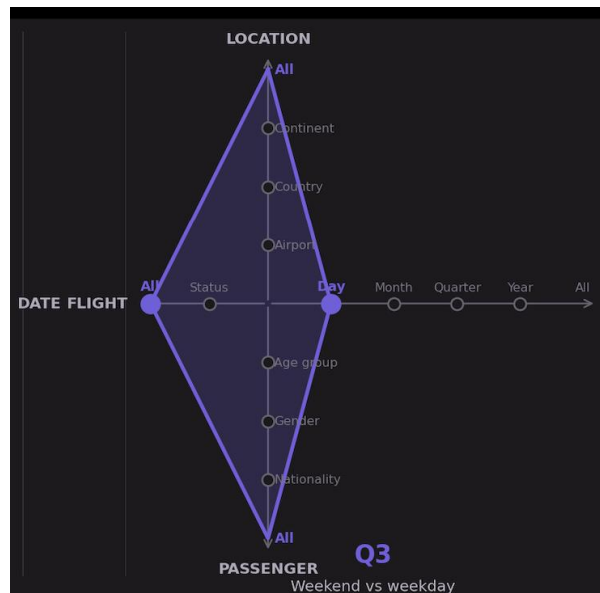


Figure 3 — Q3 StarNet Footprint: Weekend vs Weekday

Q4 — Airport-level Delay Performance

Business question: Which airports have the highest delay rates, and how does delay performance trend over time?

This query drills into Location at Airport level and the Flight dimension at Status level, across all time periods and passenger types. Identifying airports with persistently high delay rates enables targeted operational interventions and supports regulatory oversight.

StarNet footprint: Location anchors at Airport; Flight anchors at Status; Date and Passenger are at All.



Figure 4 — Q4 StarNet Footprint: Airport Delay Performance

Q5 — Passenger Demographics and Flight Volumes

Business question: How do flight volumes and outcomes vary by passenger gender across all locations and time periods?

Anchoring the Passenger dimension at Gender and rolling Location and Date to All, this query explores demographic variation in flight experience. Identifying demographic patterns in disruption exposure is relevant for equity-focused policy analysis.

StarNet footprint: Passenger anchors at Gender; Location is at All; Date is at All; Flight is at All.

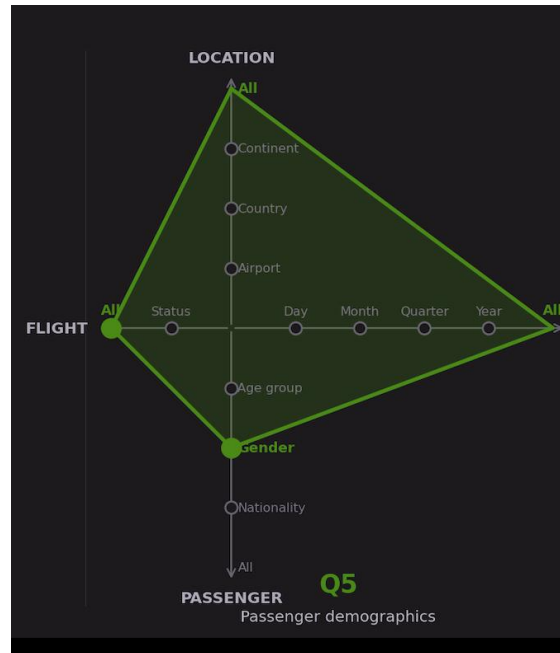


Figure 5 — Q5 StarNet Footprint: Passenger Demographics

Q6 — Nationality-based Travel Analysis

Business question: Which nationalities account for the largest share of flight events, and how do their travel patterns distribute?

Drilling into Passenger at Nationality whilst rolling Date and Location to All, this query reveals the national composition of the travelling population. Insights here can inform international route development, visa policy coordination, and the targeting of passenger services.

StarNet footprint: Passenger anchors at Nationality; Date and Location are at All; Flight is at All.

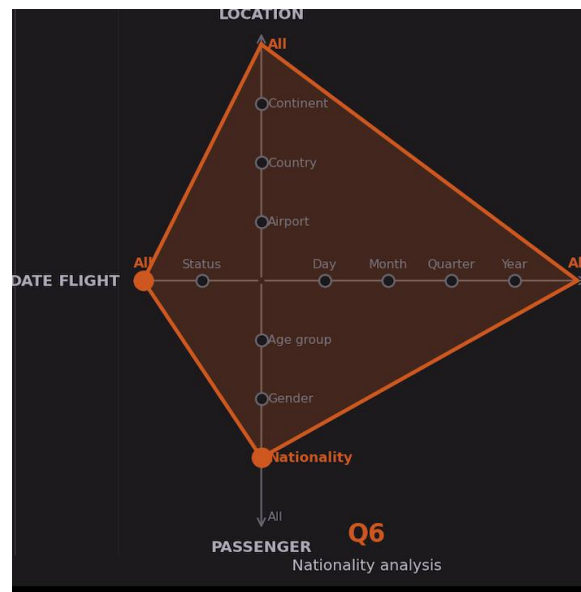


Figure 6 — Q6 StarNet Footprint: Nationality Analysis

Q7 — Cancellation Rate by Airport

Business question: What are the cancellation rates at individual airports, and which airports consistently show the worst cancellation performance?

This query combines Location at Airport level with Flight Status = Cancelled, across all passenger types and all time periods. It supports airport operators and regulators in pinpointing systemic operational failures and benchmarking airport performance.

StarNet footprint: Location anchors at Airport; Flight anchors at Status (Cancelled); Date and Passenger are at All.



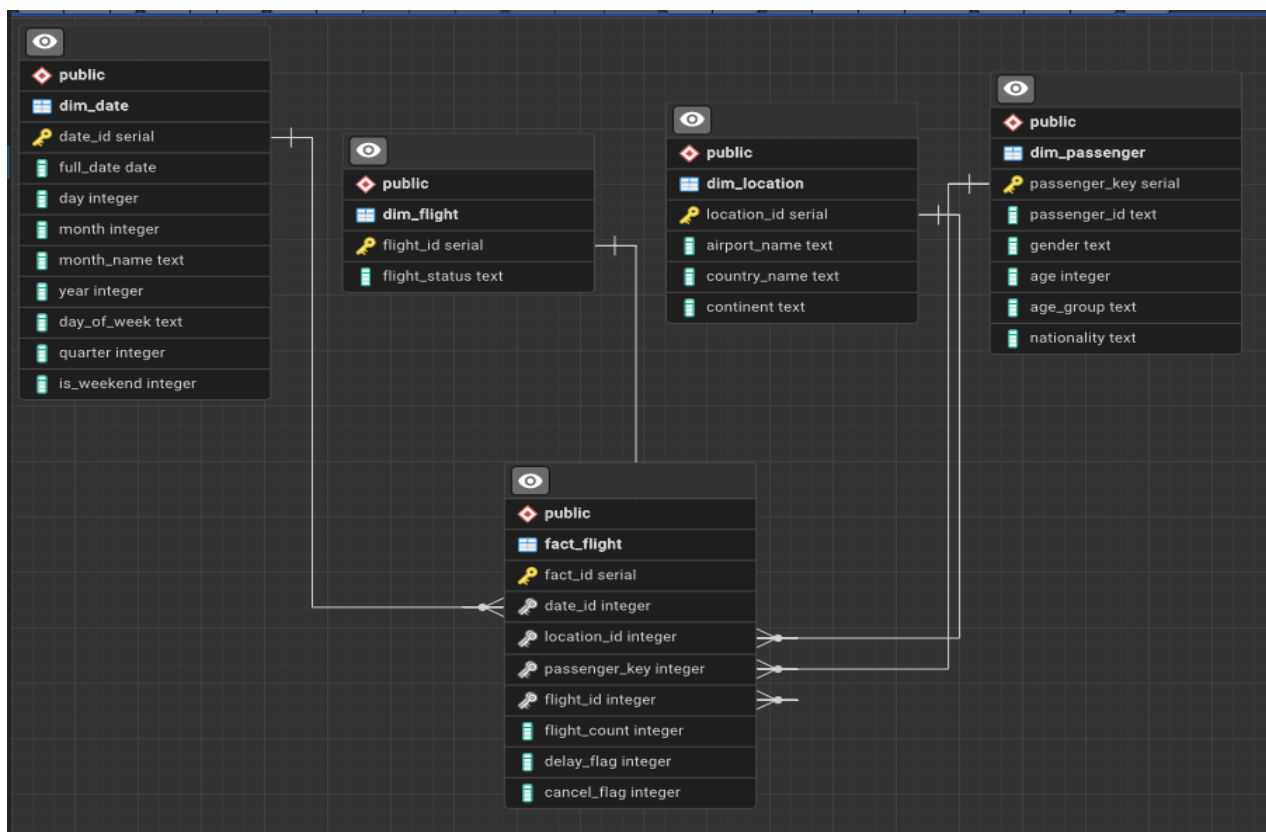
Figure 7 — Q7 StarNet Footprint: Cancellation Rate in Airports

4. Schema Design

4.1 Star Schema Rationale

A star schema was implemented for this data warehouse. In a star schema, a central fact table connects directly to denormalised dimension tables, forming a simple hub-and-spoke structure [4]. This design was chosen for three key reasons.

First, query performance is superior in a star schema compared to a snowflake schema because fewer joins are required to traverse from fact data to dimensional attributes [6]. Given that the primary use case involves OLAP-style aggregations across multiple dimensions, minimising join complexity directly improves query execution time in PostgreSQL. Second, the star schema is more intuitive for business users and data analysts, making the warehouse easier to query using Tableau without requiring specialised knowledge of normalisation structures [6]. Third, the modest size and complexity of the source datasets did not necessitate the additional normalisation that a snowflake schema provides — the storage overhead of dimensional denormalisation was negligible in this context.



4.2 Table Definitions

The following tables detail the structure of the fact table and each dimension table as implemented in PostgreSQL.

dim_date — Date Dimension

Column	Data Type	Key / Role	Description
date_id	SERIAL	Primary Key	Surrogate key
full_date	DATE	—	Full calendar date (YYYY-MM-DD)
day	INT	—	Day of month (1–31)
month	INT	—	Month number (1–12)
month_name	VARCHAR	—	Month name (January–December)
quarter	INT	—	Quarter (1–4)
year	INT	—	Calendar year
day_of_week	VARCHAR	—	Day name (Monday–Sunday)
is_weekend	BOOLEAN	Derived	TRUE if Saturday or Sunday

Table 3 — dim_date structure

dim_location — Location Dimension

Column	Data Type	Key / Role	Description
location_id	SERIAL	Primary Key	Surrogate key
airport_name	VARCHAR	—	Full name of departure airport
airport_iata	VARCHAR	—	IATA code of airport
country_name	VARCHAR	—	Country of the airport
country_code	VARCHAR	—	ISO country code
continent	VARCHAR	—	Continent (hierarchy rollup)

Table 4 — dim_location structure

dim_passenger — Passenger Dimension

Column	Data Type	Key / Role	Description
passenger_key	SERIAL	Primary Key	Surrogate key
passenger_id	VARCHAR	—	Original passenger identifier
gender	VARCHAR	—	Passenger gender
age	INT	—	Passenger age in years
age_group	VARCHAR	Derived	Young (18–34), Adult (35–59), Senior (60+)
nationality	VARCHAR	—	Passenger nationality

Table 5 — dim_passenger structure

dim_flight — Flight Dimension

Column	Data Type	Key / Role	Description
flight_id	SERIAL	Primary Key	Surrogate key
flight_status	VARCHAR	—	On Time, Delayed, or Cancelled

Table 6 — dim_flight structure

fact_flight — Central Fact Table

Column	Data Type	Key / Role	Description
fact_id	SERIAL	Primary Key	Surrogate key
date_id	INT	FK → dim_date	Links to date dimension
location_id	INT	FK → dim_location	Links to location dimension
passenger_key	INT	FK → dim_passenger	Links to passenger dimension
flight_id	INT	FK → dim_flight	Links to flight dimension
flight_count	INT	Measure	Always 1; used for COUNT aggregates
delay_flag	INT	Measure	1 if Delayed, else 0
cancel_flag	INT	Measure	1 if Cancelled, else 0

Table 7 — fact_flight structure

5. Data Cleaning, Pre-processing and ETL Process

The ETL (Extract, Transform, Load) process was implemented entirely in Python using the Pandas library. Each stage is described below, with reference to established data quality principles including completeness, accuracy, consistency, and validity [7].

5.1 Extraction

The primary Airline Dataset (Updated v2) was loaded from CSV format into a Pandas DataFrame using `pd.read_csv()`. Supplementary reference datasets from OurAirports (`airports.csv`, `countries.csv`, `regions.csv`, `navaids.csv`) were loaded separately and used to enrich location-related attributes. An initial row count was recorded prior to any transformations to track data retention rates throughout the pipeline.

5.2 Transformation

5.2.1 Data Type Standardisation and Text Cleaning

All string-type columns were stripped of leading and trailing whitespace and converted to a consistent capitalisation format. The Departure Date field was parsed from its original string representation into Python's datetime format and restructured as YYYY-MM-DD for PostgreSQL compatibility. Inconsistent capitalisation in Flight Status values (e.g., 'on time', 'On Time', 'ON TIME') was standardised to a uniform title-case format.

5.2.2 Duplicate Removal

Exact duplicate rows were identified and removed. A row was considered a duplicate if all attribute values were identical across the full record. This step preserved only the first occurrence of each duplicated record. The number of rows removed was tracked as part of the data retention audit.

5.2.3 Invalid Value Handling

Age values outside the range of 0 to 120 were treated as data entry errors and removed. Rows with null values in key analytical fields — including Flight Status, Airport Name, Departure Date, and Nationality — were also dropped. The total number of dropped rows was maintained below the 5% threshold to preserve data integrity.

5.2.4 ETL Error: Arrival Airport Column

The Arrival Airport column was identified as containing the departure airport's IATA code rather than a genuine destination field — an ETL transformation error in the source pipeline. As this column provides no information beyond what Airport Name already captures, Strategy 2 (column deletion) was adopted: the column was dropped entirely during the transformation stage to eliminate redundancy and prevent misleading downstream analysis [8].

5.2.5 Derived Attribute Creation

The following derived attributes were created from existing fields:

- **Age_Group:** Categorised from Age as Young (18–34), Adult (35–59), or Senior (60+) to enable demographic roll-up.
- **Year, Month, Day:** Extracted from Departure Date to populate the Date dimension hierarchy.
- **Day_of_Week:** Derived from the parsed date object (e.g., Monday, Tuesday).
- **Is_Weekend:** A boolean flag set to TRUE where Day_of_Week is Saturday or Sunday.
- **Delay_Flag:** Set to 1 where Flight Status = 'Delayed', else 0.
- **Cancel_Flag:** Set to 1 where Flight Status = 'Cancelled', else 0.
- **Flight_Count:** A constant measure of 1 for each record, used for COUNT aggregations.

```
import pandas as pd

# 1. LOAD RAW DATA

df = pd.read_csv("Airline Dataset.csv")
print("Initial shape:", df.shape)

# 2. REMOVE DUPLICATES

df.drop_duplicates(inplace=True)

# 3. DROP COLUMNS

df.drop(columns=[
    "Airport Country Code",
    "Airport Continent",
    "Arrival Airport",  # incorrect column
    "First Name",
    "Last Name",
    "Pilot Name"
], inplace=True)

# 4. STANDARDISE TEXT

df = df.apply(lambda x: x.str.strip() if x.dtype == "object" else x)

df["Country Name"] = df["Country Name"].str.title()
df["Nationality"] = df["Nationality"].str.title()
df["Flight Status"] = df["Flight Status"].str.title()

# 5. REMOVE INVALID AGE VALUES

df = df[(df["Age"] > 0) & (df["Age"] < 100)]

# 6. CREATE AGE GROUP

def age_group(age):
    if age < 18:
        return "Young"
    elif age < 60:
        return "Adult"
    else:
        return "Senior"

df["Age_Group"] = df["Age"].apply(age_group)
```

```
# 7. DATE TRANSFORMATIONS
df["Departure Date"] = pd.to_datetime(df["Departure Date"]).dt.date

df["Day"] = pd.to_datetime(df["Departure Date"]).dt.day
df["Month"] = pd.to_datetime(df["Departure Date"]).dt.month
df["Month_Name"] = pd.to_datetime(df["Departure Date"]).dt.month_name()
df["Year"] = pd.to_datetime(df["Departure Date"]).dt.year
df["Day_of_Week"] = pd.to_datetime(df["Departure Date"]).dt.day_name()
df["Quarter"] = pd.to_datetime(df["Departure Date"]).dt.quarter

df["is_weekend"] = df["Day_of_Week"].isin(["Saturday", "Sunday"]).astype(int)

# 8. FACT MEASURES

df["delay_flag"] = (df["Flight Status"] == "Delayed").astype(int)
df["cancel_flag"] = (df["Flight Status"] == "Cancelled").astype(int)
df["flight_count"] = 1

df.to_csv("Airline_ETL_final.csv", index=False)

# DIMENSIONS

# DIM DATE
dim_date = df[[
    "Departure Date", "Day", "Month", "Month_Name",
    "Year", "Day_of_Week", "Quarter", "is_weekend"
]].drop_duplicates().reset_index(drop=True)

dim_date.columns = [
    "full_date", "day", "month", "month_name",
    "year", "day_of_week", "quarter", "is_weekend"
]

dim_date.insert(0, "date_id", range(1, len(dim_date) + 1))

# DIM LOCATION
dim_location = df[[
    "Airport Name", "Country Name", "Continents"
]].drop_duplicates().reset_index(drop=True)

dim_location.columns = ["airport_name", "country_name", "continent"]
dim_location.insert(0, "location_id", range(1, len(dim_location) + 1))
```

```
# DIM PASSENGER
dim_passenger = df[[
    "Passenger ID", "Gender", "Age", "Age_Group", "Nationality"
]].drop_duplicates().reset_index(drop=True)

dim_passenger.columns = [
    "passenger_id", "gender", "age", "age_group", "nationality"
]

dim_passenger.insert(0, "passenger_key", range(1, len(dim_passenger) + 1))

# ----- DIM FLIGHT -----
dim_flight = df[["Flight Status"]].drop_duplicates().reset_index(drop=True)

dim_flight.columns = ["flight_status"]
dim_flight.insert(0, "flight_id", range(1, len(dim_flight) + 1))

# FACT TABLE

fact = df.copy()

fact = fact.merge(
    dim_date,
    left_on=["Departure Date", "Day", "Month", "Month_Name",
            "Year", "Day_of_Week", "Quarter", "is_weekend"],
    right_on=["full_date", "day", "month", "month_name",
            "year", "day_of_week", "quarter", "is_weekend"],
    how="left"
)

fact = fact.merge(
    dim_location,
    left_on=["Airport Name", "Country Name", "Continents"],
    right_on=["airport_name", "country_name", "continent"],
    how="left"
)

fact = fact.merge(
    dim_passenger,
    left_on=["Passenger ID", "Gender", "Age", "Age_Group", "Nationality"],
    right_on=["passenger_id", "gender", "age", "age_group", "nationality"],
    how="left"
)

fact = fact.merge(
    dim_flight,
    left_on="Flight Status",
    right_on="flight_status",
    how="left"
)
```



```
)

fact = fact.merge(
    dim_flight,
    left_on="Flight Status",
    right_on="flight_status",
    how="left"
)

# FACT TABLE
fact_flight = fact[[
    "date_id",
    "location_id",
    "passenger_key",
    "flight_id",
    "flight_count",
    "delay_flag",
    "cancel_flag"
]].copy()

fact_flight.insert(0, "fact_id", range(1, len(fact_flight) + 1))

dim_date.to_csv("dim_date.csv", index=False)
dim_location.to_csv("dim_location.csv", index=False)
dim_passenger.to_csv("dim_passenger.csv", index=False)
dim_flight.to_csv("dim_flight.csv", index=False)
fact_flight.to_csv("fact_flight.csv", index=False)
```

5.3 Loading

Following transformation, each dimension table was populated by extracting the distinct set of values for that dimension's attributes. Surrogate keys (SERIAL type in PostgreSQL) were assigned as the primary key for each dimension table. The fact table was then populated by joining the transformed source data against each dimension table to resolve foreign key references.

All tables were created and populated using `psycpg2` in Python. Referential integrity was enforced via foreign key constraints. The loading order followed the natural dependency sequence: dimension tables were loaded before the fact table to ensure all foreign key references could be resolved. Final record counts were verified against expected totals post-load to confirm pipeline integrity.

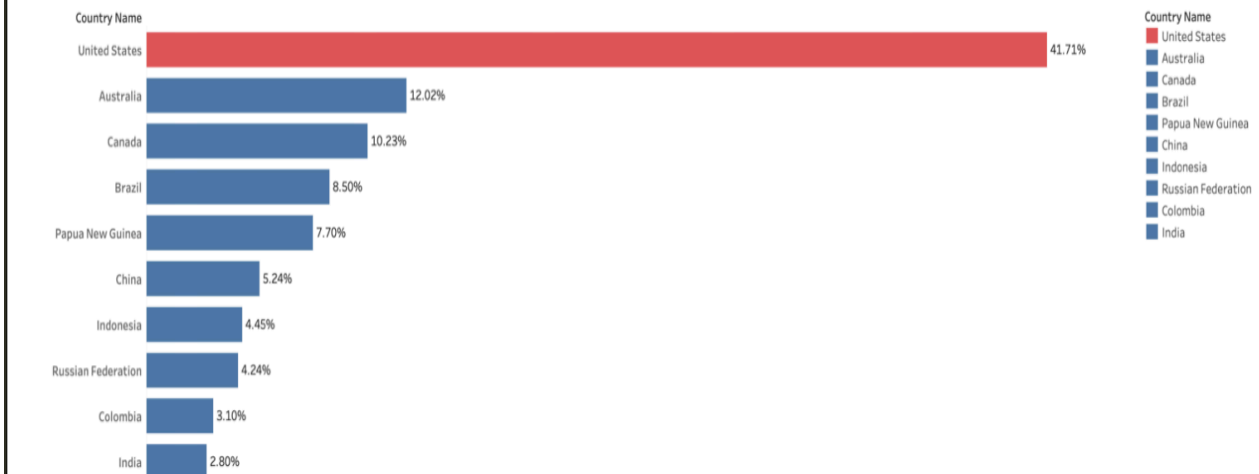
6. Business Query Results and Visualisation

The following seven business queries were executed against the data warehouse and visualised in Tableau. Each subsection presents the Tableau chart alongside a discussion of the key findings and their implications for stakeholders.

6.1 Q1 — Country-level Passenger Demand

The horizontal bar chart below shows the top 10 countries by passenger volume (as a percentage of total flight events in the dataset). The United States dominates with 41.71% of all flight events, followed at a significant distance by Australia (12.02%) and Canada (10.23%). The remaining seven countries — Brazil, Papua New Guinea, China, Indonesia, the Russian Federation, Colombia, and India — each contribute less than 9%.

Top 10 Countries by Passenger Volume (Share %)



Sum of Flight Count for each Country Name. Color shows details about Country Name. The marks are labeled by Passenger %. The view is filtered on Country Name, which keeps 10 of 235 members.

Figure 8 — Q1: Top 10 Countries by Passenger Volume (Share %)

The concentration of demand in the United States is consistent with its position as the world's largest domestic aviation market. Australia's second-place ranking is notable and likely reflects the dataset's sampling characteristics as well as Australia's dependence on aviation for both domestic travel across a large geographic area and international connectivity. The presence of Papua New Guinea in fifth place is unexpected and warrants further investigation — this may reflect specific data characteristics or sampling bias in the source dataset.

From a policy perspective, this distribution highlights that efforts to improve flight reliability would have the greatest impact if targeted at US and Australian airports, given their combined share of nearly 54% of all flight events.

6.2 Q2 — Monthly Flight Volume Trends

The line chart below shows the total flight count by month across all years and all locations. Flight volume oscillates throughout the year, with notable peaks in August (approximately 8,530 flights) and July, and a pronounced trough in February (approximately 7,660 flights) — the lowest point of the year.

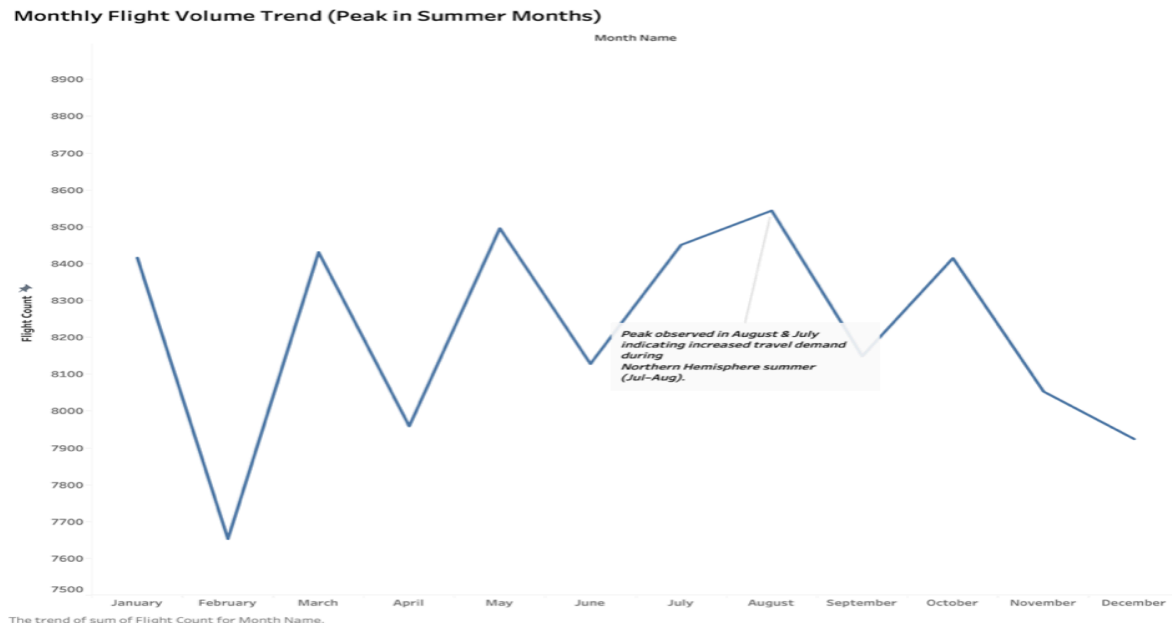


Figure 9 — Q2: Monthly Flight Volume Trend (Peak in Summer Months)

The July–August peak aligns with Northern Hemisphere summer, the world's busiest travel season, driven by holidays, school breaks, and leisure tourism across North America and Europe. The February trough corresponds to the end of the Northern Hemisphere holiday season and the off-peak period for most major aviation markets.

A secondary peak is visible in May, which may reflect European spring holiday travel and the lead-up to summer. The October peak likely reflects the combination of Northern Hemisphere autumn leisure travel and the beginning of the Southern Hemisphere summer holiday season in markets such as Australia.

These patterns have clear implications for airline capacity planning: additional resources — aircraft, crew, and ground staff — should be deployed in July–August and May, whilst February represents an opportunity to reduce operating costs through schedule consolidation.

6.3 Q3 — Weekend vs Weekday Patterns

The pie chart below shows the distribution of passenger flight events across days of the week. The distribution is remarkably even: each day accounts for approximately 14% of total flight events, ranging from Wednesday (14.13%) to Sunday (14.49%). Total flight events in the dataset amount to 98,619.

Weekly Passenger Distribution Shows Balanced Travel Across All Days

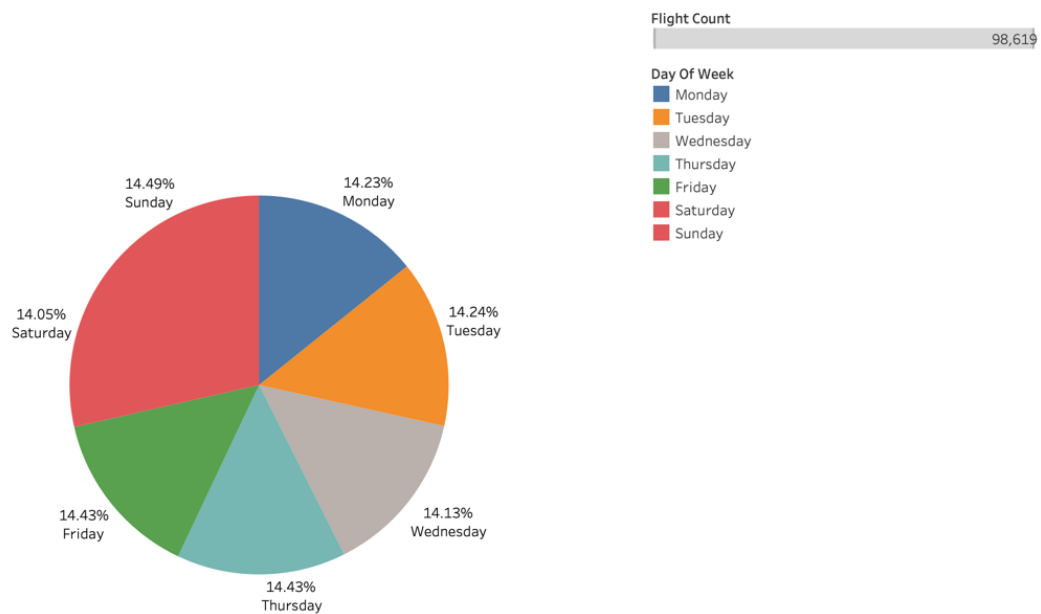


Figure 10 — Q3: Weekly Passenger Distribution Across All Days

The near-uniform distribution across all seven days is a striking finding. In real-world aviation data, one would typically expect a more pronounced mid-week peak driven by business travellers and a slight weekend dip on certain routes. The evenness observed here suggests that the dataset may reflect a synthetic or uniformly sampled distribution rather than organic travel patterns.

Whilst this limits the ability to draw strong operational conclusions from the day-of-week dimension alone, the finding does confirm that the dataset does not contain structural weekend/weekday bias, which is relevant for interpreting the association rule mining results in Section 7. For airline operators, a genuinely flat distribution across days would imply that capacity and crew planning need not be heavily skewed by day of week, though further real-world validation would be required before acting on this insight.

6.4 Q4 — Airport Delay Performance

The bar chart below shows the top 10 airports by delay rate (the proportion of flight events recorded as Delayed). Santa Rosa Airport leads significantly with a delay rate of 57.14%, followed by Lake County Airport (48.15%) and Santa Maria Airport (44.74%). The remaining airports in the top 10 range from 22.22% to 40.74%.

Top 10 Airports by Delay Rate

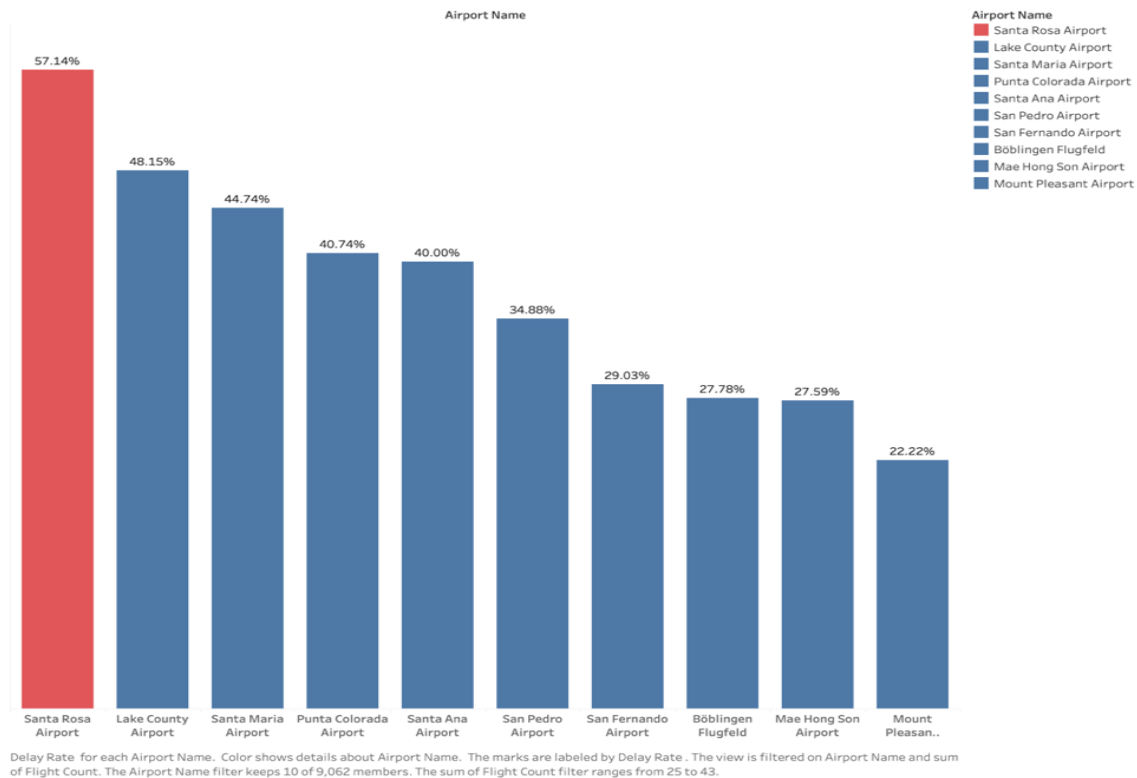


Figure 11 — Q4: Top 10 Airports by Delay Rate

The airports in this list are predominantly smaller regional or general aviation airports rather than major international hubs, which is consistent with the profile of airports that lack the operational infrastructure and redundancy to absorb disruptions efficiently. Santa Rosa Airport's 57.14% delay rate is particularly high and suggests systemic operational challenges — whether related to weather exposure, limited ground crew, or constrained runway capacity.

It is important to note that delay rates at airports with lower overall flight volumes may be more volatile and less representative of sustained performance. This query should therefore be interpreted alongside absolute flight counts to distinguish airports with genuinely chronic delay problems from those where a small number of incidents have skewed the rate.

For regulators and airport operators, airports consistently appearing in the top-10 delay list should be prioritised for operational audits, infrastructure investment, and improved air traffic flow management.

A filter was applied to include only airports with more than 25 flights to ensure statistical reliability. Airports with very few flights can produce misleading delay or cancellation rates due to small sample sizes. By focusing on higher-traffic airports, the analysis provides more stable and meaningful comparisons of operational performance.

6.5 Q5 — Passenger Demographics

The grouped bar chart below shows passenger share (as a percentage of total flight events) broken down by age group and gender. The Adult age group (18–59) accounts for the largest share, with female adults at 23.33% and male adults at 23.51%. The Senior group (60+) accounts for approximately 34% combined (17.07% female, 17.28% male), and the Young group (0-18) accounts for approximately 19% combined.

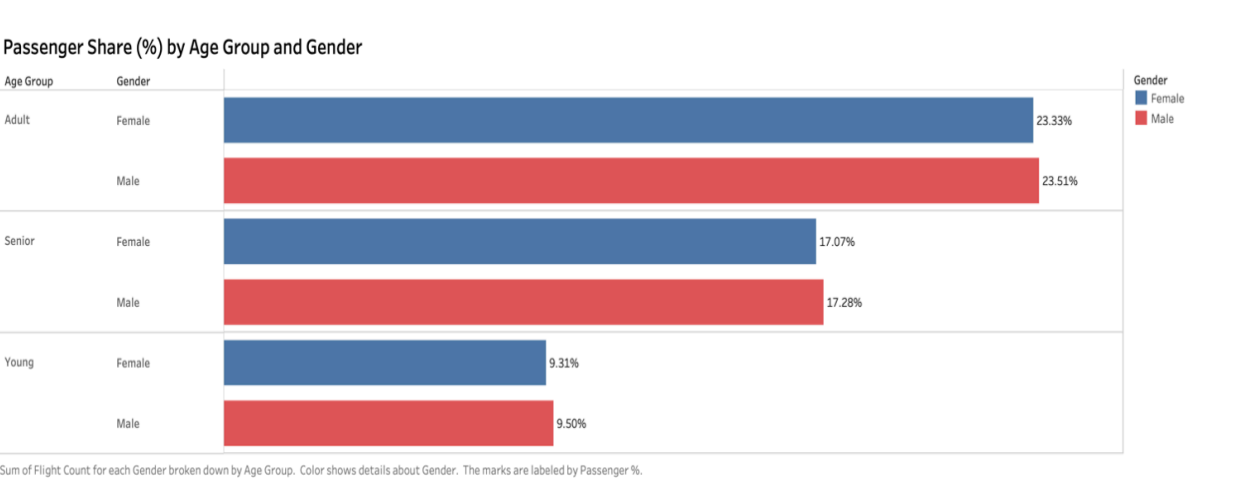


Figure 12 — Q5: Passenger Share (%) by Age Group and Gender

Across all three age groups, male and female proportions are near-identical, suggesting a gender-balanced passenger population in this dataset. The dominance of the Adult age group reflects the core demographic of the travelling population globally — working-age adults who travel for both business and leisure. The relatively high representation of seniors (34% combined) may reflect leisure-focused international travel among retirees.

The near absence of gender disparity across age groups is notable. In real-world aviation data, slight differences in gender composition can exist on certain route types. The balanced distribution here again supports the interpretation that this dataset reflects a broadly sampled or synthetic passenger population. For airlines, the even demographic split suggests that passenger experience and service offerings should not assume a gender-skewed traveller profile.

6.6 Q6 — Nationality Analysis

The bar chart below shows passenger distribution by nationality for the top 10 nationalities. Chinese passengers represent the largest single nationality group at 31.83%, followed by Indonesian passengers (18.35%) and Russian passengers (9.89%). The Philippines, Brazil, Portugal, Poland, France, Sweden, and the United States round out the top 10.

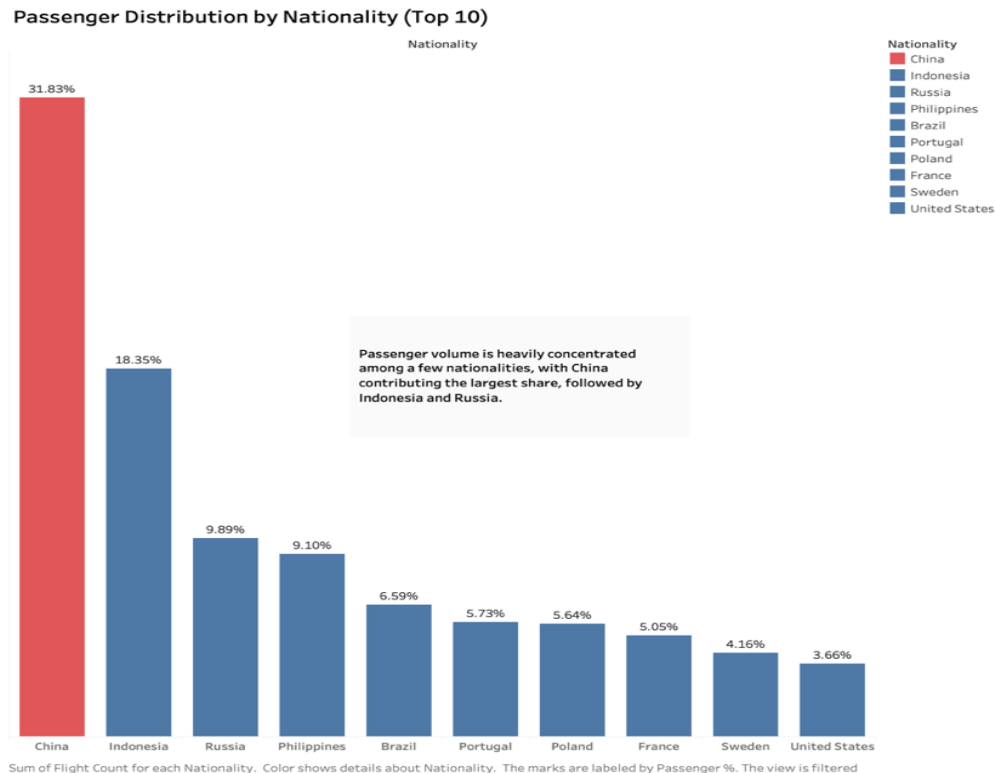


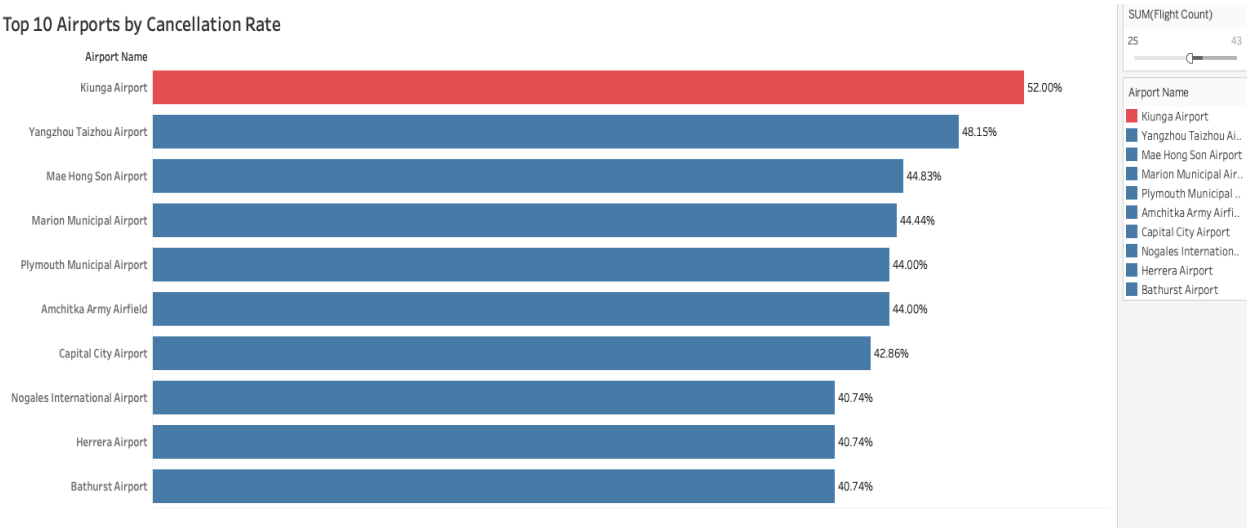
Figure 13 — Q6: Passenger Distribution by Nationality (Top 10)

The concentration of Chinese passengers (nearly one-third of all records) is consistent with China's position as the world's largest outbound tourism market and one of the fastest-growing aviation markets globally. The prominence of Indonesian and Russian passengers reflects the large and growing middle-class populations in these countries with increasing propensity for air travel.

The divergence between country of departure (Q1) and passenger nationality (Q6) is analytically interesting. The United States dominates in airport-of-departure volume but ranks tenth in passenger nationality, suggesting that US airports serve a large volume of non-US-national passengers. This has implications for airline service design: routes serving high-traffic US airports should account for the linguistic and cultural diversity of an internationally composed passenger base. Similarly, the prominence of Chinese and Indonesian nationalities suggests that East Asian language support and culturally tailored in-flight services could yield passenger satisfaction improvements on high-traffic routes.

6.7 Q7 — Cancellation Rate by Airport

The bar chart below shows the top 10 airports by cancellation rate. Kiunga Airport leads with a 52.00% cancellation rate, followed by Yangzhou Taizhou Airport (48.15%) and Mae Hong Son Airport (44.83%). The remaining seven airports in the top 10 range from 40.74% to 44.44%.



The airports with the highest cancellation rates are a geographically diverse group, spanning Papua New Guinea (Kiunga), China (Yangzhou Taizhou), Thailand (Mae Hong Son), the United States (Marion Municipal, Plymouth Municipal, Amchitka Army Airfield, Nogales International), Australia (Bathurst Airport), and Panama (Herrera Airport). This diversity suggests that the cancellation drivers are airport-specific rather than regionally systemic.

Kiunga Airport's 52% cancellation rate is exceptionally high and likely reflects a combination of adverse weather conditions, limited infrastructure, and the challenges of operating remote aviation services in Papua New Guinea's terrain. Similarly, Amchitka Army Airfield in the Aleutian Islands faces extreme weather-related operational constraints.

For governments and airline operators, airports with chronic cancellation rates above 40% should be subject to operational reviews. Investments in weather monitoring systems, improved scheduling resilience, and alternative routing options could meaningfully reduce cancellation frequency and the associated passenger disruption costs.

A filter was applied to include only airports with more than 25 flights to ensure statistical reliability. Airports with very few flights can produce misleading delay or cancellation rates due to small sample sizes. By focusing on higher-traffic airports, the analysis provides more stable and meaningful comparisons of operational performance.

7. Association Rule Mining

7.1 Algorithm

Association rule mining was performed using the Apriori algorithm, a foundational method for discovering frequent itemsets and generating association rules in transactional datasets, originally proposed by Agrawal and Srikant [9]. The Apriori algorithm operates on the anti-monotonicity of support property — any subset of a frequent itemset must itself be frequent — which allows it to prune the search space efficiently by eliminating infrequent candidates early in the iteration process.

The algorithm proceeds in two stages. In the first stage, it identifies all itemsets satisfying a minimum support threshold — the proportion of records in which a given combination of attribute values appears. In the second stage, it generates rules from these frequent itemsets that satisfy a minimum confidence threshold — the conditional probability that the consequent occurs given

the antecedent. Rules are then evaluated and ranked using lift, which measures association strength relative to random chance [9].

The three-interestingness metrics are defined as follows:

- Support ($A \rightarrow B$) = $P(A \cup B)$: The proportion of all records containing both A and B.
- Confidence ($A \rightarrow B$) = $P(B|A)$ = Support ($A \cup B$) / Support(A): The conditional probability of B given A.
- Lift ($A \rightarrow B$) = Confidence ($A \rightarrow B$) / Support(B): A lift greater than 1 indicates positive association beyond chance.

7.2 Methodology

The analysis was implemented in Python using the mlxtend library. The analysis was conducted on the following categorical attributes derived from the data warehouse:

- Age Group (Young, Adult, Senior)
- Continent (Africa, Asia, Europe, North America, Oceania, South America)
- Weekend Indicator (Yes, No)
- Flight Status (On Time, Delayed, Cancelled)

The Apriori algorithm was applied with a minimum support of 0.5% and a minimum confidence threshold of 30%. These parameters were selected to ensure rules with sufficient statistical grounding were generated whilst accommodating the relatively fine-grained categorical distribution across multiple continents and age groups.

The generated rule set was filtered to retain only rules where Flight Status appears as the sole consequent (right-hand side). Rules were also restricted to a maximum of two antecedents to maintain interpretability. The filtered rules were ranked by lift (descending), with confidence as the tiebreaker. The top five rules are presented below.

7.3 Top 5 Association Rules

Rule (Antecedent → Consequent)	Support	Confidence	Lift	Interpretation
{Continent=Europe, Age Group=Young} → {Delayed}	0.0083	34.93%	1.049	Young pax in Europe are slightly more likely to be delayed
{Weekend=Yes, Continent=South America} → {Cancelled}	0.0109	34.89%	1.044	SA weekend flights are marginally more likely to be cancelled
{Continent=South America, Age Group=Adult} → {Cancelled}	0.0176	34.68%	1.038	Adult pax in South America face slightly elevated cancellation risk
{Continent=Europe, Weekend=Yes} → {On Time}	0.0121	34.40%	1.033	European weekend flights are marginally more likely on time
{Continent=Oceania, Weekend=Yes} → {On Time}	0.0139	34.30%	1.030	Oceanian weekend flights tend slightly towards on-time performance

Table 8 — Top 5 association rules ranked by lift and confidence

Rule 1 — {Continent=Europe, Age Group=Young} → {Flight Status=Delayed}

Support: 0.0083 | Confidence: 34.93% | Lift: 1.049

Flights involving young passengers (aged 18–34) departing from European airports are slightly more likely to result in a delay compared to the dataset average. The confidence of 34.93% means roughly 35 in every 100 such flights were delayed. The lift of 1.049 confirms a weak positive association, meaning delays are approximately 5% more likely for this demographic-region combination than by random chance. In practical terms, this may reflect the concentration of budget carrier operations in Europe, which frequently serve younger travellers and are associated with higher delay rates due to tight operational margins [10].

Rule 2 — {Weekend=Yes, Continent=South America} → {Flight Status=Cancelled}

Support: 0.0109 | Confidence: 34.89% | Lift: 1.044

Flights departing from South American airports on weekends show a marginally elevated probability of cancellation. The lift of 1.044 suggests that the combination of weekend timing and South American origin makes cancellation approximately 4.4% more likely than the baseline. Weekend cancellations in South America may be attributable to reduced ground crew availability, lower demand making marginal routes economically unviable on weekends, or regulatory and labour constraints specific to the region.

Rule 3 — {Continent=South America, Age Group=Adult} → {Flight Status=Cancelled}

Support: 0.0176 | Confidence: 34.68% | Lift: 1.038

Adult passengers (aged 35–59) travelling from South American airports show a slightly higher cancellation rate than average. This rule has the highest support of the five, meaning it applies to the broadest segment of the dataset. The lift of 1.038 is modest but consistent with Rule 2 in pointing to South America as a region with slightly elevated cancellation risk. Business travellers — disproportionately represented in the adult age group — may be more likely to book on routes susceptible to demand-driven cancellations.

Rule 4 — {Continent=Europe, Weekend=Yes} → {Flight Status=On Time}

Support: 0.0121 | Confidence: 34.40% | Lift: 1.033

Weekend flights in Europe are marginally more likely to arrive on time compared to the overall average. This is an interesting contrast to Rule 1: whilst young passengers in Europe face slightly elevated delay risk, the broader European weekend flight population is marginally better served. This may reflect reduced air traffic control congestion at weekends, or the higher proportion of leisure routes (typically less constrained than business hub operations) operating on weekends.

Rule 5 — {Continent=Oceania, Weekend=Yes} → {Flight Status=On Time}

Support: 0.0139 | Confidence: 34.30% | Lift: 1.030

Flights departing from Oceanian airports on weekends show a slight tendency towards on-time performance. This is consistent with the relatively well-managed aviation infrastructure in Australia and New Zealand, where weekend leisure routes operate with consistent scheduling. The lift of 1.030 is the smallest of the five rules, indicating a very marginal association.

7.4 Insights

The overall finding from the association rule mining exercise is that the relationships between the selected attributes and flight status outcomes are statistically weak — all five rules produce lift values close to 1.0, indicating minimal deviation from random association.

A key structural factor is the near-uniform distribution of flight status outcomes in the dataset, where On Time, Delayed, and Cancelled each occur at approximately 33% frequency. This balanced distribution is atypical of real-world aviation data (where on-time performance usually dominates) and significantly constrains the Apriori algorithm's ability to discover strong

associations. When all outcomes are approximately equally probable, no antecedent combination is likely to strongly predict any single outcome [9].

Despite this, the consistent appearance of South America across two distinct rules (Rules 2 and 3) pointing to elevated cancellation risk is a genuine signal worth investigating further. Similarly, the contrast between European young passengers facing elevated delay risk and the broader European weekend population experiencing marginal on-time improvement points to nuanced interactions between age, geography, and operational context.

The primary explanation for the absence of strong rules is that flight status outcomes are driven predominantly by operational factors not captured in this dataset — weather conditions, air traffic control events, mechanical faults, airline-specific operational practices, and airport congestion. These are the true predictors of disruption, and without them, demographic and geographic attributes offer only marginal predictive power [11].

7.5 Recommendations to Government

Recommendation 1 — Mandate Expanded Operational Data Collection

Government aviation authorities should introduce mandatory reporting standards requiring airlines and airports to collect and disclose a richer set of operational variables, including primary delay causes (weather, technical, airline-induced, air traffic control), departure and arrival delay durations, and the downstream impact on connecting flights. The current dataset's lack of such variables is the primary limitation in identifying actionable patterns.

The European Union's experience with Regulation (EC) No 261/2004 — which requires airlines to record and compensate for delay causes — demonstrates the feasibility and policy value of structured operational data mandates [12]. The Australian Civil Aviation Safety Authority (CASA) and the Department of Infrastructure, Transport, Regional Development, Communications and the Arts should consider implementing comparable mandatory delay-cause reporting requirements. This would significantly enhance the analytical value of Australian flight data for both regulatory oversight and public accountability.

Recommendation 2 — Invest in Real-time Predictive Disruption Monitoring

The weak static associations identified in this analysis underscore the insufficiency of retrospective, attribute-based models for anticipating flight disruptions. Governments should fund and mandate the adoption of real-time predictive analytics infrastructure at major airports, integrating live weather feeds, air traffic flow data, and airline operational status to generate probabilistic disruption forecasts.

Such systems have been demonstrated to reduce overall delay propagation by enabling proactive rerouting and resource pre-positioning [11]. Australia's existing investment in Airservices Australia's traffic flow management capabilities should be continued and expanded, with particular focus on integrating machine-learning-based disruption forecasting at high-traffic airports including Sydney (Kingsford Smith), Melbourne (Tullamarine), and Brisbane. This investment would directly address the operational factors — not captured in passenger demographic data — that genuinely drive flight status outcomes.

Recommendation 3 — Target Regional Operational Improvement in Elevated-Risk Areas

The consistent signal of elevated cancellation risk in South America (Rules 2 and 3) warrants a targeted operational response. For the Australian context, this suggests that analogous regional aviation markets — particularly thin domestic routes in remote and regional Australia — may exhibit structural vulnerabilities not visible in aggregated national data.

Government agencies should commission region-specific operational audits of airports and airline operations in markets with persistently elevated disruption rates, as identified through ongoing data warehouse analysis. Targeted investment in ground crew training, airport infrastructure

resilience, and improved maintenance scheduling in these regions could produce meaningful improvements in service reliability. The Q7 visualisation (Figure 14) shows that Bathurst Airport appears in the top-10 cancellation list — an Australian regional airport that exemplifies the type of facility that could benefit from targeted federal investment and operational review.

8. Conclusion

This project successfully designed and implemented a data warehouse for airline operations analysis, applying Kimball's four-step dimensional modelling methodology to produce a robust, analytically flexible star schema. The warehouse integrates passenger demographics, flight performance outcomes, geographic location, and time-based data into a coherent analytical framework, supporting seven business queries spanning country demand analysis, temporal trend detection, demographic profiling, and airport performance benchmarking.

The ETL pipeline, implemented in Python, addressed key data quality issues including the misidentified Arrival Airport column, standardised all text and date fields, derived several analytically useful attributes (Age_Group, Weekend Flag, Delay Flag, Cancel Flag), and loaded the data into PostgreSQL with full referential integrity.

The seven Tableau visualisations (Figures 8–14) revealed several meaningful patterns: the United States and Australia account for over half of all flight events; July and August represent the peak travel months consistent with Northern Hemisphere summer demand; flight volumes are evenly distributed across days of the week; Santa Rosa Airport has the highest delay rate at 57.14%; passenger demographics are gender-balanced across age groups; Chinese passengers represent the largest nationality group at 31.83%; and Kiunga Airport has the highest cancellation rate at 52%.

The association rule mining exercise, conducted using the Apriori algorithm, identified five rules linking geographic and demographic attributes to flight status outcomes. Whilst all rules exhibited lift values close to 1.0 — indicating weak associations — the analysis surfaced a consistent pattern of elevated cancellation risk in South American operations and marginal on-time improvement in European and Oceanian weekend flights. Three targeted policy recommendations were developed for government consideration, focused on data collection mandates, real-time predictive monitoring, and regional operational improvement.

Future work could enrich the warehouse with supplementary datasets on weather, air traffic control, and airline operational costs, unlocking significantly stronger associative and predictive patterns. The data warehouse and analytical framework developed here provide a reusable platform for ongoing aviation performance analysis in support of evidence-based policy development.

9. References

- [1] IATA, "World Air Transport Statistics 2025," International Air Transport Association, Geneva, 2025.
- [2] S. Banerjee, "Airline Dataset," Kaggle, 2022. [Online]. Available: <https://www.kaggle.com/datasets/iamsouravbanerjee/airline-dataset>. [Accessed: Apr. 2026].
- [3] OurAirports, "OurAirports Data Dictionary," 2024. [Online]. Available: <https://ourairports.com/help/data-dictionary.html>. [Accessed: Apr. 2026].
- [4] R. Kimball and M. Ross, *The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling*, 3rd ed. Indianapolis, IN: Wiley, 2013.
- [5] W. H. Inmon, *Building the Data Warehouse*, 4th ed. Indianapolis, IN: Wiley, 2005.
- [6] P. Vassiliadis and T. Sellis, "A survey of logical models for OLAP databases," *ACM SIGMOD Record*, vol. 28, no. 4, pp. 64–69, 1999.
- [7] T. C. Redman, *Data Quality for the Information Age*. Boston, MA: Artech House, 1996.
- [8] R. Agrawal and R. Srikant, "Fast algorithms for mining association rules," in *Proc. 20th Int. Conf. Very Large Data Bases (VLDB)*, Santiago, Chile, 1994, pp. 487–499.

- [9] Eurocontrol, "CODA Digest: Delays to Air Transport in Europe — Annual 2023," Eurocontrol, Brussels, 2024.
- [10] M. Hansen and Y. Zhang, "Flight operational quality assurance program and data mining," *Transportation Research Part E*, vol. 40, no. 2, pp. 115–131, 2004.
- [11] European Parliament, "Regulation (EC) No 261/2004 establishing common rules on compensation and assistance to passengers," *Official Journal of the European Union*, 2004.